LOW-CODE AI | CHAPTER 4

Use AutoML to Predict Advertising Media Channel Sales

Follow a regression project from marketing budget question to exploratory analysis, managed training, evaluation, feature attribution, prediction, and budget decisions.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 71–112. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Define sales outcome → Explore channel data → Train AutoML regression → Check holdout metrics → Use predictions cautiously

Detailed Notes

Business objective, data, and exploration

Source: book pp. 71–85 / PDF p. 87

The use case asks how advertising-channel spend relates to sales for a solar-energy business. Identify the label, row unit, spend units, time period, and whether future sales are genuinely predicted or historical sales are explained.

Load CSV into Pandas, check schema/nulls/duplicates, summarize distributions, and plot relationships. Scatterplots and correlations reveal patterns but not causation. Check outliers and whether channel budgets or totals leak information unavailable at prediction time. Use consistent train/validation/test partitions.

Linear regression and AutoML workflow

Source: book pp. 85–97 / PDF p. 101

Linear regression predicts a numeric target from a weighted feature combination. Its simple form supports interpretation but assumes an approximately linear relationship under the chosen feature representation.

> $\hat{y} = \beta_0 + \Sigma_j \beta_j x_j$

Upload a managed dataset, identify the target, choose regression, configure a split/training budget, and train. Verify the service's split behavior, data types, and feature transformations. AutoML may compare multiple pipelines, but it does not validate the business framing.

Evaluate, interpret, predict, decide

Source: book pp. 98–111 / PDF p. 114

Use MAE for average absolute error in target units, RMSE when large errors should be penalized more, and R-squared as variance-relative fit—not as a guarantee of prediction quality. Compare to a naive baseline and inspect residuals by channel, spend range, and time.

Feature importance/attribution describes model reliance under a particular method and dataset; it is not causal proof that increasing spend will produce the attributed sales. Use predictions for scenario planning only within supported data ranges. Confirm assumptions with experiments or domain evidence before reallocating budget.

Revision Check

1. Is the task prediction or causal budget allocation?
2. What baseline and residual slices matter?
3. How do MAE, RMSE, and R-squared differ?
4. Why is feature attribution not causal impact?

A regression score is only useful when its target, validation split, and intended decision match the business question.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.